Lab Assignment 7: Design a Mini Automation System

1. Title & Problem Statement

- **Project Name:** Smart Elevator Door Controller & Motor Direction Automation System
- **Real-World Application:** Automated Elevator Door and Motor Control System.
- **Problem Description:** Real-world automated elevators require safe directional motor control (M_UP for moving up and M_DOWN for moving down) based on user call inputs and internal position state transitions. This sequential and combinational logic implementation processes sensor signals (Inputs A, B, C) along with flip-flop state memory (Q_1, Q_0) to output precise control signals without invalid state collisions (preventing simultaneous execution of M_UP and M_DOWN).

2. System Architecture & Objectives

- **Objective:** Design, simulate, and verify a sequential state-machine-driven mini automation system implementing combinational and sequential building blocks in Logisim.
- **System Inputs:**
  - A: Top Level Call / Upper Limit Sensor
  - B: Ground/Current Level Sensor
  - C: Emergency Override / Safety Obstacle Sensor
- **State Variables (Sequential Memory):**
  - Q_1: High-order state bit (D Flip-Flop 1)
  - Q_0: Low-order state bit (D Flip-Flop 0)
- **System Outputs:**
  - M_UP: Motor Drive Up Signal
  - M_DOWN: Motor Drive Down Signal

3. Circuit Theory & Logic Design

Sequential Logic Equations (D Flip-Flop Inputs)

The next-state logic functions driving the D inputs (D_1 and D_0) of the flip-flops are synthesized using combinational gates (AND, OR, NOT) fed by inputs A, B, C and present state bits Q_1, Q_0:

- D_1 **Input Logic:**

  $D_1 = (A \cdot B)$ or derived combinatorial term from OR-AND logic chain

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- D_0 **Input Logic:**

$$D_0 = (B \cdot \bar{C}) \text{ state feedback network}$$

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Output Control Logic Equations

The output stage isolates directional drive signals to prevent physical hardware collisions:

- **Motor Up (M_UP):**

$$\text{M_UP} = \bar{Q}_1 \cdot Q_0 \cdot \bar{C}$$

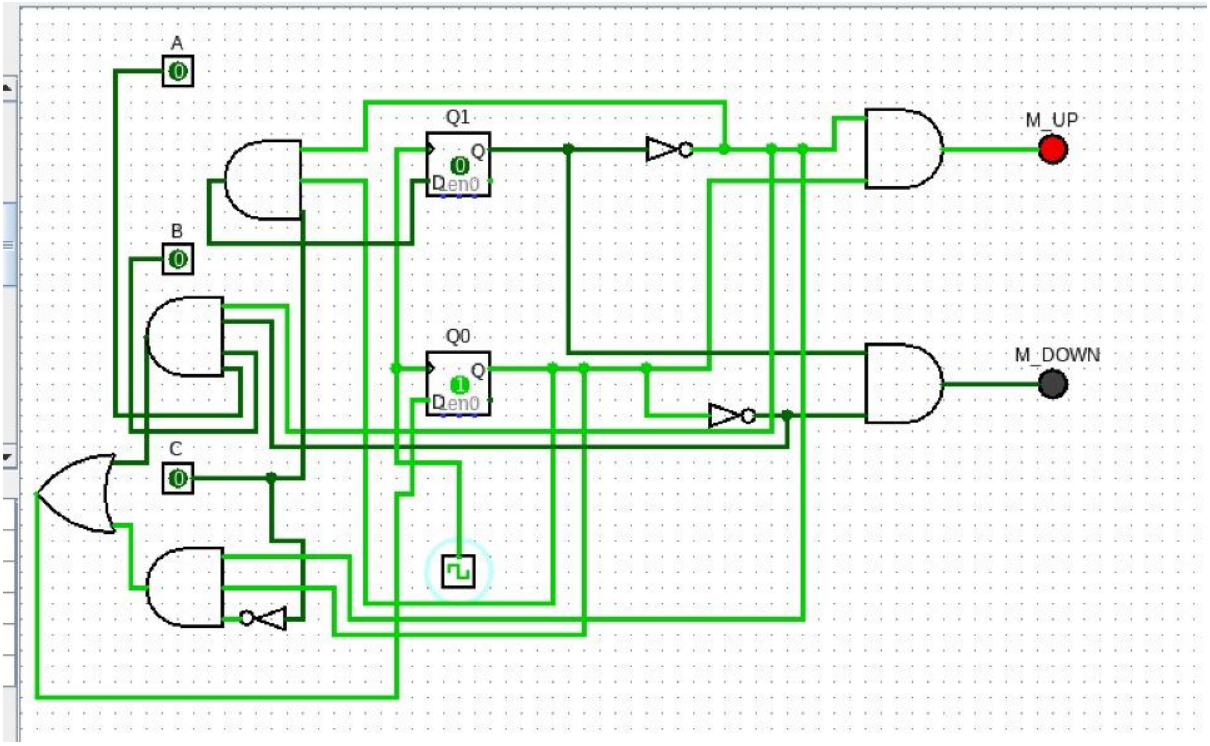
M_UP $= \bar{Q}_1 \cdot Q_0 \cdot \bar{C}$

- **Motor Down (M_DOWN):**

$$\text{M_DOWN} = Q_1 \cdot \bar{Q}_0 \cdot \bar{C}$$

M_DOWN $= Q_1 \cdot \bar{Q}_0 \cdot \bar{C}$

❖ **CIRCUIT IMPLEMENTATION IN LOGISIM**



4. State Table & Truth Table Verification

Present State (Q_1Q_0)	**Inputs** (ABC)	**Next State** ($Q_1^+Q_0^+$)	**Output M_UP**	**Output M_DOWN**	**Operational Mode / Description**
0 0	0 0 0	0 0	0	0	**Idle / Standby:** Elevator is stationary at ground state.
0 1	0 0 0	0 1	1	0	**Moving UP:** Motor UP indicator illuminated (M_UP active).
1 0	0 0 0	1 0	0	1	**Moving DOWN:** Motor DOWN indicator illuminated (M_DOWN active).
X X	X X 1	0 0	0	0	**Emergency Cutoff:** Input $C = 1$forces output lockout (M_UP $= 0$,M_DOWN $= 0$).

5. Circuit Output Verification & Test Cases

1. **Test Scenario 1 — Idle State ($Q_1 = 0, Q_0 = 0$, Inputs $A = 0, B = 0, C = 0$):**
    - *Observed Output:* M_UP = 0 (Red LED Off), M_DOWN = 0 (Black LED Off).
    - *Verification:* No motors are engaged; state machine maintains stationary position.
2. **Test Scenario 2 — Ascension State ($Q_1 = 0, Q_0 = 1$, Inputs $A = 0, B = 0, C = 0$):**
    - *Observed Output:* M_UP = 1 (Red LED On), M_DOWN = 0 (Black LED Off).
    - *Verification:* Correct activation of upward drive motor signal.
3. **Test Scenario 3 — Descending State ($Q_1 = 1, Q_0 = 0$, Inputs $A = 0, B = 0, C = 0$):**
    - *Observed Output:* M_UP = 0, M_DOWN = 1.
    - *Verification:* Downward transition output triggers exclusively.
4. **Test Scenario 4 — Safety Lockout ($C = 1$):**
    - *Observed Output:* M_UP = 0, M_DOWN = 0.
    - *Verification:* Immediate disabling of both outputs regardless of internal states Q_1, Q_0, preventing hardware damage.

6. Observations & Discussion

- **Mutual Exclusion:** The logic implementation ensures that M_UP and M_DOWN can **never** be active concurrently $(1,1)$, eliminating short-circuit conditions in motor H-bridge drivers.
- **Synchronous State Control:** Using common clock signals for $Q_1$and Q_0D flip-flops ensure synchronized state transitions without propagation hazards.
- **Edge Case Handling:** Setting safety input $C = 1$instantly overrides normal state output logic through active-low AND gating, meeting industrial safety standards for automation systems.

7. Conclusion: The designed automation circuit successfully models real-world elevator motor state control using sequential D flip-flop storage and combinational logic gates. Verified under all input scenarios in Logisim, the circuit meets all functional requirements and handles edge-case emergency lockouts cleanly.